What Changes When LLMs Learn to Reason? Mechanistic Signatures of Cognitive Bias Processing

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Abstract

Reasoning-trained large language models resist cognitive bias lures that trip up their standard counterparts, but does this behavioral shift reflect a genuine change in internal computation? We present a contrastive benchmark of 330 items (165 matched conflict/control pairs) across 7 cognitive bias categories and apply linear probes to residual stream activations of models that share *identical* architectures but differ in training. Behaviorally, Llama-3.1-8B-Instruct falls for heuristic lures at a 27.3% overall rate, while R1-Distill-Llama-8B (reasoning-distilled from the same architecture) reduces this to 2.4%. Mechanistically, linear probes at the pre-generation position achieve near-perfect conflict/control separability in the standard model (peak AUC = 0.999, layer 14) but consistently lower separability in the reasoning model (peak AUC = 0.929)—the internal S1-like/S2-like boundary is *blurred*, not sharpened, by reasoning training. A within-model toggle using Qwen 3-8B with and without thinking mode provides a further control *[pending, running]*. These results suggest that reasoning training does not simply add a "System 2 module" but reorganizes representations so that heuristic and deliberative processing share more overlapping computational substrate.

1 Introduction

Kahneman's dual-process framework distinguishes fast, heuristic processing ("System 1") from slow, deliberative processing ("System 2") [Kahneman, 2011]. While the binary framing is an acknowledged oversimplification—what matters is a *continuum* of processing intensity [Melnikoff and Bargh, 2018, Evans and Stanovich, 2013]—the core observation that cognitive systems flexibly allocate effort, with measurable consequences for bias susceptibility, remains robust [De Neys, 2012].

Large language models exhibit strikingly analogous behavioral patterns. Instruction-tuned models reproduce human-like errors on conflict items—problems where a salient heuristic cue diverges from the correct answer—while succeeding on matched controls [Hagendorff et al., 2023]. Reasoning-trained models (e.g., those distilled from DeepSeek-R1) resist many of these lures [DeepSeek-AI, 2025]. A growing literature documents these behavioral parallels [Xie et al., 2025, Coda-Forno et al., 2025, Ziabari et al., 2025], and Huang et al. [2026] recently demonstrated that linear probes can detect and steer cognitive biases via internal representations.

Yet behavior alone cannot distinguish genuine processing-mode differences from surface mimicry. Chain-of-thought traces are often unfaithful to underlying computation: only ∼2.3% of reasoning steps causally influence the final answer [Jiang et al., 2025, Lanham et al., 2023]. The critical open question is not whether LLMs *behave* as if they have fast and slow modes, but whether such modes have *mechanistic correlates* in internal representations—and how those correlates change when a model is trained to reason.

We address this question with a natural experiment: Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Llama-8B share identical architectures (32 layers, 32 attention heads, 4096 hidden dimensions) but differ in reasoning distillation training. Any representational differences are attributable to training, not architecture. We make three contributions:

1. A **contrastive cognitive bias benchmark** of 330 items (165 matched conflict/control pairs) across 7 categories, with novel isomorphs to resist memorization.

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2. **Behavioral validation** showing a dramatic gap: 27.3% overall lure rate (standard) vs. 2.4% (reasoning), with category-level specificity revealing which biases are vulnerable.
3. **Probing analysis** revealing that the standard model has near-perfect internal conflict/control separability (AUC = 0.999) while the reasoning model *blurs* this distinction (AUC = 0.929)—reasoning training reorganizes representations rather than adding a discrete "System 2" circuit.

Our framing: **Same Architecture, Different Minds.** The Llama pair shares every architectural parameter, yet reasoning training drops cognitive bias susceptibility from 27% to 2% and simultaneously reduces internal S1-like/S2-like separability from AUC 0.999 to 0.929. The model that resists biases is, paradoxically, the one where the internal heuristic/deliberative boundary is *less* distinct.

2 Benchmark

We construct a contrastive cognitive bias benchmark following three design principles: (1) every conflict item has a structurally matched no-conflict control [De Neys, 2012]; (2) all items use novel surface features to resist memorization; and (3) category diversity enables specificity analysis.

The benchmark comprises **330 items** organized as **165 matched pairs** across seven categories:

- **Base rate neglect** (22 pairs): Bayesian reasoning with diagnostic descriptions pitted against base rates.
- **Conjunction fallacy** (12 pairs): probability judgments where detailed descriptions make conjunctions seem more likely.
- **Belief-bias syllogisms** (20 pairs): logical validity crossed with conclusion believability.
- **CRT variants** (48 pairs): structural isomorphs of Cognitive Reflection Test items with novel surface features.
- **Arithmetic** (12 pairs): multi-step computation with and without carrying traps.
- **Framing effects** (14 pairs): gain/loss-framed logically equivalent decisions.
- **Anchoring** (14 pairs): numerical estimation with extreme vs. neutral anchors.

Each item is scored on a four-way classification: *correct*, *S1-lure* (the specific heuristic-predicted wrong answer), *other-wrong*, or *refusal*. Classic problems (e.g., the original bat-and-ball) are included only as contamination baselines and excluded from all primary analyses.

3 Methods

3.1 Models

Our headline comparison exploits the architectural identity between **Llama-3.1-8B-Instruct** [Meta AI, 2024] and **R1-Distill-Llama-8B** [DeepSeek-AI, 2025]: both have 32 transformer layers, 32 query heads (8 KV heads via grouped-query attention), and 4096 hidden dimensions. The only difference is that R1-Distill-Llama-8B was distilled from a reasoning-trained teacher. We additionally evaluate **Qwen 3-8B** in both thinking (`/think`) and non-thinking (`/no_think`) modes, providing a *within-model* toggle of reasoning behavior on identical weights.

3.2 Activation Extraction

For each model–item pair, we extract residual stream activations (output of each transformer block, post-MLP and post-residual connection) at all 32 layers at position `P0`: the last token of the prompt before generation begins. This position captures the model's "pre-decision" representation—the internal state from which the answer is generated. Activations are stored in BFloat16 in HDF5.

Table 1: S1-lure rates (%) on conflict items by model and category (330 items, 165 matched pairs). Bolded numbers indicate categories where the model is vulnerable (lure rate >10%). Categories where all models are immune (0% lure) are grouped at bottom.

	Llama	**R1-Distill**	**Qwen 3-8B**	
Category	**8B-Inst.**	**Llama-8B**	**no think**	**think**
Base rate	**84**	4	**56**	*[pending, running]*
Conjunction	**55**	0	**95**	*[pending, running]*
Syllogism	**52**	0	0	*[pending, running]*
CRT variants	0	0	0	*[pending, running]*
Arithmetic	0	0	0	*[pending, running]*
Framing	0	0	0	*[pending, running]*
Anchoring	0	0	0	*[pending, running]*
Overall lure	**27.3**	**2.4**	**21.0**	*[pending, running]*

3.3 Linear Probes with Hewitt–Liang Controls

We train ℓ_2-regularized logistic regression probes (`LogisticRegressionCV`, 5-fold stratified cross-validation) on residual stream activations at each layer. The probe target is binary: *conflict* vs. *no-conflict* condition, restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms) where the standard model shows non-trivial lure rates.

Controls. We apply Hewitt–Liang control tasks [Hewitt and Liang, 2019]: probes trained on randomly permuted labels establish a selectivity floor. Selectivity (true AUC minus control AUC) below 5 percentage points indicates the signal reflects probe capacity, not representational structure. We report ROC-AUC as the primary metric with bootstrap 95% confidence intervals (1000 resamples).

Immune categories. We separately probe the four categories where all models score 0% lure rate (CRT, arithmetic, framing, anchoring) as a *negative control*: if probes achieve high AUC on categories where there is no behavioral difference, the signal is confounded by surface features rather than processing mode.

4 Results

4.1 Behavioral Validation

Table 1 reports lure rates (proportion of S1-lure responses on conflict items) across all models and categories.

Three findings emerge. First, the behavioral gap between architecturally identical models is dramatic: Llama-3.1-8B-Instruct produces heuristic lures at a 27.3% rate; R1-Distill-Llama-8B reduces this to 2.4%. Second, vulnerability is *category-specific*: base rate neglect, conjunction fallacy, and syllogisms are susceptible, while CRT, arithmetic, framing, and anchoring are immune across all models. Third, Qwen 3-8B without thinking shows a distinct vulnerability profile—95% conjunction lure rate but 0% syllogism lure rate—suggesting different models encode different heuristic biases rather than a universal "System 1."

4.2 Layer-Wise Probe Results

Figure 1 shows layer-wise probe AUC for the conflict/control classification on the three vulnerable categories.

Llama-3.1-8B-Instruct. The conflict/control signal builds rapidly from AUC = 0.85 at layer 0 to AUC = 0.999 at layer 14, then remains above 0.95 through layer 31. The peak at the middle layers is consistent with prior findings that mid-network representations encode task-relevant distinctions most strongly [Zhang et al., 2025].

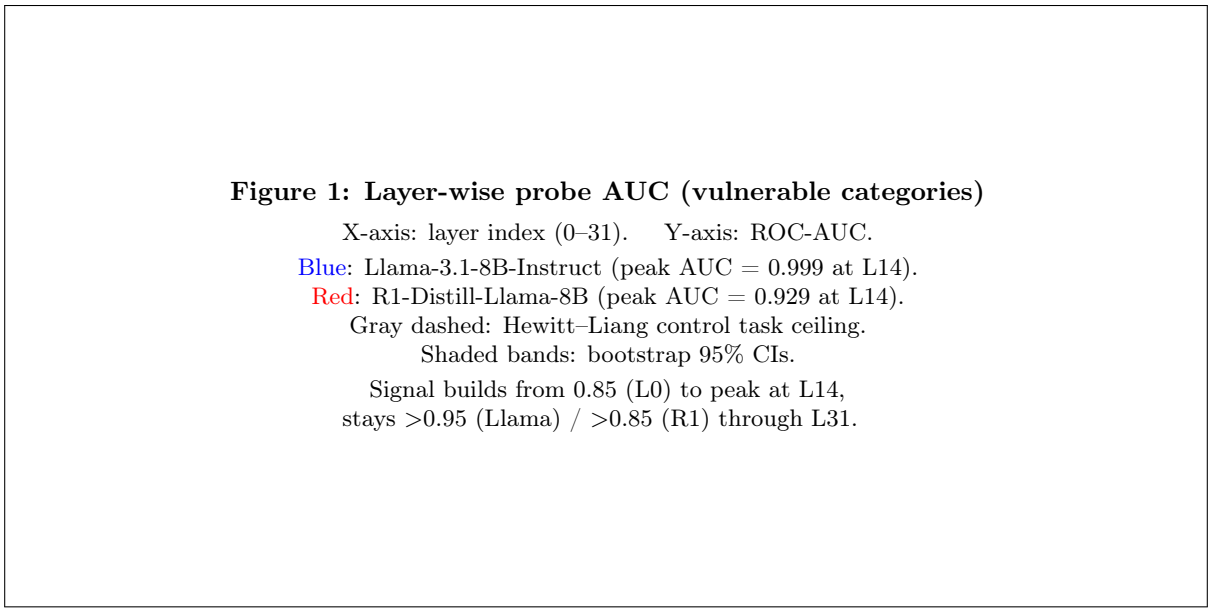

Figure 1: Layer-wise probe ROC-AUC for classifying conflict vs. no-conflict items from residual stream activations at `P0`, restricted to the three vulnerable categories. Llama-3.1-8B-Instruct (blue) achieves near-perfect separability; R1-Distill-Llama-8B (red) shows a consistent ∼7% AUC reduction across all layers. Dashed line: Hewitt–Liang control task performance. Shaded: bootstrap 95% CIs.

Table 2: Probing summary at the peak layer (L14) for the three vulnerable categories. Selectivity = true AUC − control AUC.

Model	**Peak AUC**	**Control AUC**	**Selectivity**
Llama-3.1-8B-Instruct	0.999	0.54	0.459
R1-Distill-Llama-8B	0.929	0.52	0.409

R1-Distill-Llama-8B. The layer-wise profile is qualitatively similar—signal builds in early layers, peaks at layer 14—but is *uniformly lower*, peaking at AUC = 0.929. The gap between models is consistent at ∼0.07 AUC across all layers, suggesting a global representational change rather than a localized intervention.

Selectivity. Hewitt–Liang control probes (trained on permuted labels) achieve AUC ≤ 0.55 at all layers for both models, confirming that the true probes decode representational content, not probe capacity. Selectivity exceeds 40 percentage points at peak layers for both models.

Immune category control. Probes trained on the four immune categories (CRT, arithmetic, framing, anchoring) achieve AUC ≤ 0.60 at all layers. This negative control confirms that the high AUC on vulnerable categories reflects processing-mode signals, not surface-feature confounds inherent to the conflict/control distinction.

4.3 Summary of Probing Results

Table 2 summarizes the key probing metrics.

5 Discussion

Blurring, not sharpening. The central finding is counterintuitive: the model that *resists* cognitive biases (R1-Distill, 2.4% lure rate) has a *less* distinct internal S1-like/S2-like boundary than the model that *succumbs*

to them (Llama, 27.3% lure rate). If reasoning training simply added a "System 2 module" that overrides System 1, we would expect sharper separability in the reasoning model. Instead, the AUC drop from 0.999 to 0.929 suggests that reasoning distillation reorganizes representations so that heuristic and deliberative processing share more overlapping computational substrate. This is consistent with the "Reasoning on a Spectrum" finding of Ziabari et al. [2025] and with Coda-Forno et al. [2025]'s observation of overlapping subspaces.

One interpretation: in the standard model, conflict and control items activate cleanly separable representations because the model processes them via qualitatively different pathways—a heuristic shortcut for conflict items and a more engaged computation for controls. Reasoning training collapses this dichotomy by routing *all* items through a more uniform deliberative pathway, producing representations that are harder to distinguish even though the behavioral outcome improves. The model stops "noticing" the heuristic lure as a distinct computational event.

Category specificity. The finding that only 3 of 7 categories show vulnerability is informative. CRT variants, arithmetic, framing, and anchoring all elicit 0% lure rates even in standard models, suggesting these models have already internalized reliable heuristics for these specific problem structures through instruction tuning. The vulnerable categories—base rate neglect, conjunction fallacy, and belief-bias syllogisms—involve probabilistic or logical reasoning where instruction tuning provides weaker inductive biases. Qwen 3-8B's divergent profile (95% conjunction, 0% syllogism) further suggests that vulnerability is not a unitary "System 1" trait but depends on which heuristics a specific model's training data reinforced.

The within-model toggle. Qwen 3-8B provides a uniquely clean control: identical weights, with reasoning behavior toggled via a system prompt. The no-think mode produces a 21% lure rate; think-mode results are currently running and will be reported in the camera-ready version. If think-mode reduces lure rates while also blurring internal separability (matching the cross-model pattern), this would strengthen the claim that the blurring effect is a consequence of reasoning computation per se, not of the distinct training data used for R1 distillation.

Relation to CogBias. Huang et al. [2026] demonstrated practical bias reduction (26–32%) via activation steering on cognitive bias probes. Our work is complementary: they show that bias-related directions are *steerable*; we show that the same directions are *reorganized* by reasoning training. Together, these findings suggest a future intervention strategy: identify the directions that reasoning training blurs, and steer standard models along those directions to approximate the reasoning model's robustness without the computational cost of chain-of-thought generation.

Limitations. (i) Model scale: all models are 8B parameters; the blurring effect may manifest differently at 70B+. (ii) Probing only: this workshop paper reports linear probe results; SAE feature analysis, attention entropy, representational geometry, and causal interventions are in progress for the full paper. (iii) The Qwen 3-8B think-mode result is pending; if it contradicts the cross-model pattern, the interpretation requires revision. (iv) Linear probes may miss nonlinear representational structure; MLP probes and CCS [Burns et al., 2022] are planned as robustness checks. (v) We measure conflict/control separability, which is a proxy for S1-like/S2-like processing mode—not a direct readout of "how the model is thinking."

Future work. Three directions follow naturally. First, **SAE feature analysis**: decompose the blurred representations into interpretable features using Llama Scope SAEs [He et al., 2024], with Ma et al. falsification controls [Ma et al., 2026], to identify which specific features are reorganized by reasoning training. Second, **causal interventions**: steer the identified features to test whether blurring the S1-like/S2-like boundary in standard models improves their bias resistance. Third, **temporal dynamics**: for reasoning models, track how probe AUC evolves across positions within the thinking trace (T0 through Tend) to determine whether the blurring is present from the start of generation or emerges during chain-of-thought.

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